

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 20

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• Most sulfur in coal is present as (solid) iron pyrites (<math>\text{FeS}_2</math>)</li> <li>• Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming)</li> </ul>                                                                                                                                                                                                                                                  | 2     |
| 01.2     | <ul style="list-style-type: none"> <li>• Flue gases are passed through a bed of crushed calcium carbonate (limestone)</li> <li>• Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide)</li> <li>• If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster</li> </ul>                                                                                                                                                | 3     |
| 01.3     | <ul style="list-style-type: none"> <li>• The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide</li> <li>• The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource</li> </ul>                                                                                                                                                                                                                               | 2     |
| 01.4     | <ul style="list-style-type: none"> <li>• Fluidised bed: increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form <math>\text{NO}_x</math> (production prevented)</li> <li>• Catalytic converter: platinum / palladium catalyst chemically reduces <math>\text{NO}_x</math> back to nitrogen and oxygen gases in the exhaust</li> <li>• Urea sprayed into the hot flue / exhaust gases reacts with <math>\text{NO}_x</math>, converting it to nitrogen, carbon dioxide and water</li> </ul> | 3     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 02.1     | <ul style="list-style-type: none"> <li>• B - Steam washing (hot water / steam jets kill organisms in the sediment)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
| 02.2     | <ul style="list-style-type: none"> <li>• Chemicals cause the oil molecules to join together / solidify</li> <li>• Producing more solid material that can be collected / lifted more easily than liquid oil</li> </ul>                                                                                                                                                                                                                                                                                                                            | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.3     | <ul style="list-style-type: none"> <li>• Air contains about <math>210 \text{ cm}^3</math> of oxygen per litre whereas water at <math>20^\circ \text{C}</math> holds a maximum of about <math>6 \text{ cm}^3</math> per litre (less than 3% of the amount in air)</li> <li>• So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance</li> </ul>                                                                                | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| 03.1     | <ul style="list-style-type: none"> <li>• 2020 has higher peak flow in the wet season (<math>32 \text{ vs } 22 \text{ m}^3 \text{ s}^{-1}</math> in March)</li> <li>• 2020 has much lower dry-season flow (<math>2 \text{ vs } 9 \text{ m}^3 \text{ s}^{-1}</math> in July / August)</li> <li>• Flow is much more variable / seasonal in 2020; the river almost dries up</li> </ul>                                                                                                                                                                                    | 3     |
| 03.2     | <ul style="list-style-type: none"> <li>• In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow</li> <li>• With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk</li> <li>• Little stored ice remains to melt in the dry season so flow falls very low</li> </ul>                                                                                                                                         | 3     |
| 03.3     | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Water shortages for irrigation / drinking / HEP in the dry season</li> <li>• Flooding in the wet season</li> <li>• Sudden floods from ice-lake outbursts (glacial lake outburst floods)</li> </ul>                                                                                                                                                                                                                                                                                                 | 2     |
| 03.4     | <ul style="list-style-type: none"> <li>• Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances</li> <li>• But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming</li> </ul>                                                                                                                                                                                                         | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| 04.1     | <ul style="list-style-type: none"> <li>• Gasification: coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane)</li> <li>• Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS</li> <li>• Liquefaction: coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion</li> <li>• Provides liquid fuels for vehicles / aircraft that solid coal cannot supply</li> </ul> | 4     |
| 04.2     | <ul style="list-style-type: none"> <li>• A solid, ice-like crystalline structure in which methane is trapped within water ice</li> <li>• Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins</li> </ul>                                                                                                                                                                                                                                                                                             | 2     |
| 04.3     | <ul style="list-style-type: none"> <li>• Water heating: hot water pumped into sediments melts the hydrate releasing methane</li> <li>• Depressurisation: drilling reduces pressure so methane dissociates from the crystals</li> <li>• CO<sub>2</sub> injection: CO<sub>2</sub> bonds more strongly to the ice and displaces methane (could double as CCS)</li> <li>• Risks: methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback; seabed ecology poorly understood; sediment instability / landslides</li> </ul>            | 4     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 05.1     | <ul style="list-style-type: none"> <li>• No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow)</li> <li>• Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas</li> <li>• Small scale matches local demand; electricity used locally without long transmission lines / grid connection</li> <li>• Low construction cost and simple technology maintained by the community</li> <li>• Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed; Kaplan turbines with adjustable blades cope with variable flow</li> </ul> | 5     |
| 05.2     | <ul style="list-style-type: none"> <li>• Lack of capital / finance, technical knowledge or grid access; seasonal drying of rivers; conflicts with irrigation water; planning / land ownership</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| 05.3     | <ul style="list-style-type: none"> <li>• Output is proportional to flow x efficiency: <math>100 \times 0.10 \times (40 / 90)</math></li> <li>• = 4.4 kW (accept 4.4-4.5)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 05.4     | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost; high flows that clear gravel spawning beds lost; floodplain no longer flooded</li> <li>• Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis</li> <li>• Sediment trapped so less nutrient replenishment downstream; barrier to migrating fish (salmon, sturgeon) and isolation of populations</li> </ul>                                                                                                                                                                                        | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| 06.1     | <ul style="list-style-type: none"> <li>• There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change</li> <li>• Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
| 06.2     | <ul style="list-style-type: none"> <li>• Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs</li> <li>• Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises</li> <li>• Calculate standard deviation to assess scatter: more variable data need more samples</li> <li>• Test the method (extraction technique, timing) for reliability before the main study</li> </ul>                                                                                                                                    | 4     |
| 06.3     | <ul style="list-style-type: none"> <li>• Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates</li> <li>• Collect worms that come to the surface for a standardised time; count / identify / weigh</li> <li>• Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed</li> </ul>                                                                                                                                                                                                                      | 4     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 07.1     | <ul style="list-style-type: none"> <li>• Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines</li> <li>• Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost</li> </ul>                                                                                           | 2     |
| 07.2     | <ul style="list-style-type: none"> <li>• As volume increases the surface area : volume ratio decreases</li> <li>• Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage)</li> </ul>                                                                                                                                                          | 2     |
| 07.3     | <ul style="list-style-type: none"> <li>• Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation</li> <li>• Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines</li> </ul>                                                                                                                                               | 2     |
| 07.4     | <ul style="list-style-type: none"> <li>• Materials with high specific heat capacity (concrete, water) warm and cool slowly</li> <li>• They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating</li> </ul>                                                                                                                                                                | 2     |
| 07.5     | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation</li> <li>• Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage</li> <li>• Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries</li> </ul>                         | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                     |       |
| 08.1     | <ul style="list-style-type: none"> <li>• Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel)</li> <li>• Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably</li> <li>• Wood / biofuels are renewable but depletable if forests are felled faster than they regrow</li> </ul>  | 3     |
| 08.2     | <ul style="list-style-type: none"> <li>• Other factors (technology, location, cost) restrict how much can be harnessed</li> <li>• Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures</li> <li>• New technologies or higher prices may make these usable in future</li> </ul> | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
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| <b>08.3</b> | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid</li> <li>• HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock</li> <li>• Solar: high light intensity, low cloud cover</li> <li>• Wave: coasts with strong reliable prevailing winds and a long fetch</li> <li>• Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow</li> <li>• Geothermal: hot rocks near the surface / recent volcanic activity</li> <li>• Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel</li> </ul> | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |
| <b>09.1</b> | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up</li> <li>• High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation)</li> <li>• Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification)</li> <li>• Birds of prey are at the top of the food chain so receive the highest concentrations</li> <li>• Sub-lethal doses caused thin eggshells that broke / reduced breeding success; higher doses caused sterility or death</li> </ul>                                                                                                                                  | 4     |
| <b>09.2</b> | <ul style="list-style-type: none"> <li>• Advantage: low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly</li> <li>• Disadvantage: high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase); chronic effects (memory loss, depression); suspected carcinogens</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <b>09.3</b> | <ul style="list-style-type: none"> <li>• Pyrethroids are highly toxic to fish (even though mammal toxicity is low)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| <b>09.4</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic)</li> <li>• They are moderately persistent so remain in the soil long enough to be absorbed</li> <li>• Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats</li> <li>• So bees / pollinators are less exposed than when the crop is sprayed</li> </ul>                                                                                                                                                                                                                                                                                                                     | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
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| 10.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs</li> <li>• Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling</li> <li>• Mechanisation allowing deeper underground mines and larger open-cast mines</li> <li>• Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater</li> <li>• Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues</li> <li>• Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy</li> <li>• Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion</li> </ul> | 9     |
| 10.2     | <ul style="list-style-type: none"> <li>• All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 130-147; AQA spec 3.2.3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |

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| 11.1     | <p><b>Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use</li> <li>• Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles</li> <li>• Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour</li> <li>• Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management</li> <li>• Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour</li> <li>• Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| 11.2     | <p><b>Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Composition and structure: N<sub>2</sub>, O<sub>2</sub>, Ar, CO<sub>2</sub>, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry</li> <li>• Energy processes: UV absorbed by O / O<sub>2</sub> / O<sub>3</sub> dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo</li> <li>• Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements</li> <li>• Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time</li> <li>• Other threats: aerosols, smoke, land-use change; interconnection of processes</li> <li>• Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 86-113; AQA spec 3.2.1.</p> | 25    |